

Problem Setup

September 23, 2026

Problem Setup

We are given the quadratic equation

$$x^2 + 3x + 0 = 1$$

and are asked to find all real numbers x that satisfy it.

Strategy and Key Principles

The equation is a standard quadratic of the form $ax^2 + bx + c = 0$. We will bring all terms to one side to obtain the canonical form and then apply the **quadratic formula**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

which yields the exact roots of any quadratic equation.

Detailed Solution

First move the constant term on the right-hand side to the left:

$$x^2 + 3x - 1 = 0.$$

Identify the coefficients:

$$a = 1, \quad b = 3, \quad c = -1.$$

Insert these into the quadratic formula:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot (-1)}}{2 \cdot 1} \\ &= \frac{-3 \pm \sqrt{9 + 4}}{2} \\ &= \frac{-3 \pm \sqrt{13}}{2}. \end{aligned}$$

Thus the equation has two real solutions:

$$x_1 = \frac{-3 + \sqrt{13}}{2}, \quad x_2 = \frac{-3 - \sqrt{13}}{2}.$$

Final Answer

$$x = \frac{-3 \pm \sqrt{13}}{2}$$

Key Insights

- Always rewrite a quadratic equation in the form $ax^2 + bx + c = 0$ before applying the quadratic formula.
- The discriminant $b^2 - 4ac$ determines the nature of the roots; here it is positive (13), giving two distinct real solutions.
- A common mistake is to forget to move the constant term to the left side, which would lead to an incorrect constant c in the formula.